EXP.NO:15

IMPLEMENTATION OF QUICK SORT

PROBLEM STATEMENT:

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

CODE

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        int min_idx=i;
        for(int j=i+1;j<n;j++){
            if(a[j]<a[min_idx]){
                min_idx=j;
            }
        }
        int temp=a[i];
        a[i]=a[min_idx];
        a[min_idx]=temp;
    }
    for(int i=0;i<n;i++){
        printf("%d ",a[i]);
    }
}
```

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OUTPUT:



	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

RESULT:

The code is executed successfully.